

Immunodeficiency 79 (IMD79): A Comprehensive Disease Characteristics Report

Germline TET2 deficiency — an autosomal-recessive epigenetic inborn error of immunity

Summary

Immunodeficiency 79 (IMD79; OMIM #619223) is an ultra-rare Mendelian inborn error of immunity caused by **biallelic germline loss-of-function mutations in *TET2*** (ten-eleven translocation methylcytosine dioxygenase 2; HGNC:25941; locus 4q24). *TET2* is an Fe(II)/ α -ketoglutarate-dependent dioxygenase that catalyzes the iterative oxidation of 5-methylcytosine (5mC) to 5-hydroxymethylcytosine (5hmC), driving active DNA demethylation. When both alleles are lost, the enzyme is absent or catalytically dead, producing genome-wide DNA hypermethylation that dysregulates the developmental and immune transcriptional programs of hematopoietic cells. The disease was first defined in 2020 in three unrelated children carrying rare homozygous germline missense or nonsense *TET2* variants [PMID: 32518946].

Clinically, IMD79 is a **combined immunodeficiency with autoimmune lymphoproliferative syndrome (ALPS)-like features and a striking predisposition to lymphoma**. Affected children present with susceptibility to infection, chronic lymphadenopathy, hepatosplenomegaly, developmental delay, autoimmunity, and B-cell or T-cell lymphoma. The immunophenotype includes expanded double-negative (TCR $\alpha\beta$ ⁺ CD4[−]CD8[−]) T cells, depleted circulating follicular helper T cells, impaired Fas-dependent apoptosis, and defective B-cell class-switch recombination. The entity has since been expanded by additional biallelic/compound-heterozygous cases with ALPS-like disease and hematologic malignancy [PMID: 36066697], and even by heterozygous carriers presenting with B-cell lymphoma [PMID: 40031954].

Mechanistically, two arms converge to produce disease: (1) a **catalytic/epigenetic arm** in which loss of 5hmC-mediated demethylation deranges hematopoietic stem cell (HSC) self-renewal, myeloid skewing, regulatory T-cell (Treg) stability, and malignant transformation; and

(2) a **methylation-independent inflammatory arm** in which TET2 normally recruits HDAC2 to actively repress *IL6*, so that its loss unleashes IL-6-driven inflammation. Allogeneic hematopoietic stem cell transplantation (HSCT) is curative for the hematopoietic-immune disease and was achieved in all three index patients. Vitamin C (ascorbate), a cofactor of the Fe(II)/ α -KG TET enzymes, is a mechanistically rational but clinically unproven adjunct that could restore residual/paralog TET activity in hypomorphic alleles.

1. Disease Information

Overview. IMD79 is an autosomal-recessive immune dysregulation syndrome caused by biallelic germline loss-of-function of the epigenetic regulator *TET2*. It is best conceptualized as an **epigenetic combined immunodeficiency** that bridges immunodeficiency, autoimmunity/lymphoproliferation, and cancer predisposition. It represents the constitutional (germline) counterpart of the somatic *TET2* mutations long known to drive clonal hematopoiesis and myeloid/lymphoid malignancy.

Key identifiers.

Resource	Identifier
OMIM (phenotype)	#619223 — "Immunodeficiency 79"
Gene	<i>TET2</i> (OMIM *612839)
HGNC	HGNC:25941
Cytogenetic locus	4q24
MONDO	(suggested) MONDO term for "immunodeficiency 79"; map to OMIM:619223
ICD-11	4A00-region "Primary immunodeficiencies" (no dedicated code)
MeSH	No dedicated descriptor; index under <i>TET2</i> , "Immunologic Deficiency Syndromes," "Autoimmune Lymphoproliferative Syndrome"
Orphanet	No dedicated ORPHAcode identified for this ultra-rare entity as of writing

Synonyms / alternative names. Germline TET2 deficiency; autosomal-recessive germline TET2 deficiency; TET2-related childhood immunodeficiency and lymphoma; ALPS-like syndrome due to TET2 loss-of-function.

Source of information. The disease is defined almost entirely from **individual patient case series** (aggregated across a handful of families worldwide) combined with **model-organism and in vitro mechanistic data**, rather than from population-level EHR or registry resources. This is expected for an ultra-rare Mendelian condition.

2. Etiology

Disease causal factors. The primary cause is **genetic**: biallelic (homozygous or compound-heterozygous) germline loss-of-function mutations in *TET2*. Whole-exome sequencing of three unrelated affected children identified rare homozygous germline missense or nonsense *TET2* variants, with mutated TET2 protein "absent or enzymatically defective for 5-hydroxymethylating activity, resulting in whole-blood DNA hypermethylation" [PMID: 32518946]. This is a monogenic, cell-intrinsic etiology; infections in patients are consequences of the immunodeficiency rather than causal.

Genetic risk factors. The causal variants are the biallelic *TET2* LOF alleles themselves. Consanguinity contributes to homozygosity (autosomal-recessive inheritance). Heterozygous LOF variants confer a *milder / partial* risk — four heterozygous carriers developed B-cell lymphoma against a background of chronic lymphadenopathy and autoimmune features, "expand[ing] the association of germline TET2 mutations with lymphoma and an autoimmune lymphoproliferative syndrome-like phenotype to the heterozygous state" [PMID: 40031954]. Somatic "second-hit" mutations (e.g., in *DNMT3A*, *ASXL1*, or oncogenic drivers) are inferred to be required for progression to overt malignancy, consistent with TET2's role as a pre-malignant "gatekeeper."

Environmental risk factors. No specific environmental triggers are established. Age (childhood onset), and the accumulated antigenic/inflammatory burden over time, plausibly modulate the timing of lymphoproliferation and malignancy, but no toxin, occupational, or lifestyle exposure has been linked. Family history / consanguinity is the principal non-genetic risk marker.

Protective factors. No validated genetic or environmental protective factors are established. Mechanistically, **residual TET paralog activity (TET1/TET3)** and cofactor availability (Fe(II), α -ketoglutarate, ascorbate) could partially buffer hypomorphic alleles, but this is inferred rather than demonstrated in patients.

Gene–environment interactions. Inflammatory/infectious stimuli interact with the genetic lesion: because TET2 normally restrains *IL6* during resolution of inflammation, infectious/endotoxic challenges are predicted to produce exaggerated, poorly resolving inflammation in TET2-deficient individuals [PMID: 26287468]. This is a genotype (TET2 loss) $\times$ environment (inflammatory stimulus) interaction.

3. Phenotypes

The core clinical phenotype, per the defining series of three children, comprised "susceptibility to infection, lymphadenopathy, hepatosplenomegaly, developmental delay, autoimmunity, and lymphoma of B-cell (n = 2) or T-cell (n = 1) origin" [PMID: 32518946].

Phenotype	Type	HPO suggestion	Onset	Severity / frequency
Recurrent/susceptibility to infection	Clinical sign (immunodeficiency)	HP:0002721 (Immunodeficiency) / HP:0002719 (Recurrent infections)	Childhood	Variable; present in index cases
Lymphadenopathy	Physical manifestation	HP:0002716	Childhood, chronic	Common/consistent
Hepatosplenomegaly	Physical manifestation	HP:0001433	Childhood	Common
Splenomegaly	Physical manifestation	HP:0001744	Childhood	Common
Developmental delay	Clinical sign	HP:0001263	Early childhood	Present in index cases
Autoimmunity (multi-system)	Clinical sign	HP:0002960	Childhood	Prominent
Autoimmune cytopenias	Laboratory/clinical	HP:0001973 (Autoimmune thrombocytopenia) / HP:0001937	Childhood	Reported (ALPS-like)
B-cell or T-cell lymphoma	Physical manifestation	HP:0002665 (Lymphoma)	Childhood	3/3 index; major morbidity
Expanded double-negative T cells (TCRαβ+CD4–CD8–)	Laboratory abnormality	ALPS biomarker	Childhood	Characteristic
Depleted circulating follicular helper T cells	Laboratory abnormality	—	Childhood	Characteristic
Impaired Fas-dependent apoptosis	Laboratory abnormality	HP:0005404 (Abnormal lymphocyte apoptosis)	Childhood	2 of 3 patients

Phenotype	Type	HPO suggestion	Onset	Severity / frequency
Defective B-cell class-switch recombination	Laboratory abnormality	HP:0005387 (Abnormal immunoglobulin level)	Childhood	Characteristic
Elevated vitamin B12	Laboratory abnormality	— (ALPS biomarker)	Childhood	Reported [PMID: 36066697]
Elevated IL-10	Laboratory abnormality	— (ALPS biomarker)	Childhood	Reported [PMID: 36066697]
Hypergammaglobulinemia	Laboratory abnormality	HP:0010702	Childhood	Reported [PMID: 36066697]

Immunophenotype detail. "Circulating T cells showed an abnormal immunophenotype including expanded double-negative, but depleted follicular helper, T-cell compartments and impaired Fas-dependent apoptosis in 2 of 3 patients. Moreover, TET2-deficient B cells showed defective class-switch recombination" [PMID: 32518946]. The second series added "inverted myeloid/plasmacytoid dendritic cells ratio, elevated terminally differentiated effector memory CD8⁺ T-cells re-expressing CD45RA, regulatory T-cells, and Th2 circulating follicular T-cells. Double-negative T-cells, vitamin B12, and IL-10 were elevated according to the ALPS-like suspicion" [PMID: 36066697].

Age of onset / severity / progression. Onset is in **childhood**; the course is **chronic and progressive**, punctuated by lymphoproliferative episodes and culminating in lymphoma. Severity is high given lymphoma predisposition and multi-system autoimmunity.

Quality-of-life impact. Not formally measured with EQ-5D/SF-36/PROMIS in this ultra-rare cohort. Qualitatively, the combination of recurrent infection, chronic lymphoproliferation, developmental delay, autoimmune complications, malignancy, and the need for HSCT implies substantial impairment of daily functioning and major treatment burden.

4. Genetic / Molecular Information

Causal gene. *TET2* (ten-eleven translocation methylcytosine dioxygenase 2), HGNC:25941, OMIM *612839, cytoband 4q24. The protein is an Fe(II)/ α -ketoglutarate-dependent dioxygenase that oxidizes 5mC $\rightarrow$ 5hmC $\rightarrow$ 5-formylcytosine $\rightarrow$ 5-carboxylcytosine, enabling active DNA demethylation [PMID: 42423046, 42609618].

Pathogenic variants. - **Variant types:** rare homozygous germline **missense or nonsense** variants in the defining series [PMID: 32518946]; compound-heterozygous and monoallelic LOF variants in subsequent reports [PMID: 36066697, 40031954]. - **Functional consequence: loss of function** — TET2 protein "absent or enzymatically defective for 5-hydroxymethylating activity," producing whole-blood DNA hypermethylation [PMID: 32518946]. P1 in the second series showed "absent TET2 expression and profound increase in DNA methylation" [PMID: 36066697]. - **Classification (ACMG/AMP):** Nonsense/frameshift LOF alleles in a gene with established LOF disease mechanism generally reach **pathogenic/likely pathogenic**; novel missense variants require functional confirmation (loss of 5hmC activity provides strong functional evidence, PS3). - **Allele frequency:** Biallelic germline LOF is ultra-rare; causal variants are private/rare in gnomAD. (Note: *somatic* TET2 mutations are common in age-related clonal hematopoiesis and myeloid neoplasia, but these are distinct from the germline disease.) - **Somatic vs germline:** IMD79 is **germline**; the same gene is a frequent **somatic** driver in hematologic malignancy — an important distinction for genetic counseling.

Modifier genes. Inferred second-hit somatic drivers (e.g., *DNMT3A*, *ASXL1*, oncogenes) modify progression to malignancy. TET paralogs *TET1/TET3* may modify severity by partial compensation; notably TET1 is largely dispensable for MLL-ENL myeloid leukemogenesis, indicating non-redundant TET member roles [PMID: 33705482]. Regulatory partners of TET2 — the **TOPD complex (TET-OGT-PROSER1-DBHS)** — provide additional layers that could modify function [PMID: 42423046].

Epigenetic information. The disease is fundamentally epigenetic: loss of TET2 causes **genome-wide 5mC hypermethylation and reduced 5hmC**, silencing genes normally kept demethylated/active. This is the central molecular readout [PMID: 32518946, 36066697].

Chromosomal abnormalities. None characteristic at the germline level; the lesion is a point mutation/small indel in *TET2*. Secondary cytogenetic changes may accompany malignant transformation.

5. Environmental Information

Environmental factors. No specific toxin, radiation, or occupational exposure is established as causal or contributory. Disease is monogenic.

Lifestyle factors. None established.

Infectious agents. No infectious agent causes IMD79. Infections are **downstream consequences** of the immunodeficiency. Inflammatory/infectious stimuli, however, act as *triggers* that unmask the exaggerated, poorly resolving inflammation caused by loss of TET2-mediated *IL6* repression [PMID: 26287468]. In model organisms, Tet2-deficient mice "were more susceptible to endotoxin shock and dextran-sulfate-sodium-induced colitis" [PMID: 26287468], illustrating the interaction between environmental inflammatory challenge and the genetic defect.

6. Mechanism / Pathophysiology

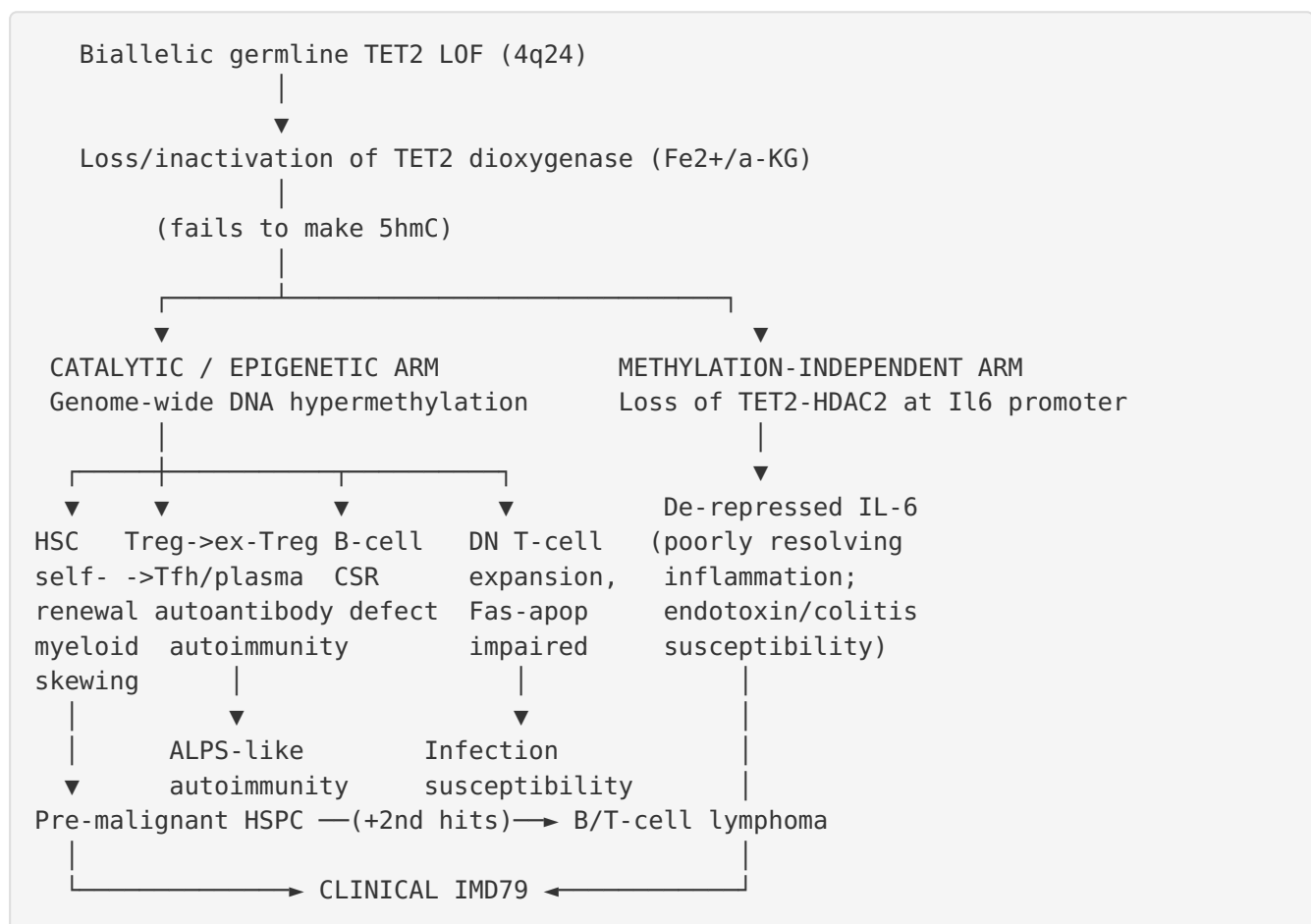
Ordered causal chain

1. **Biallelic germline LOF mutation in *TET2*** (missense/nonsense) → **loss of, or catalytically dead, TET2 protein** [PMID: 32518946].
2. Loss of TET2 dioxygenase activity → **failure to oxidize 5mC to 5hmC** → **genome-wide DNA hypermethylation / reduced 5hmC** in blood cells [PMID: 32518946, 36066697].
3. Aberrant DNA methylation → **dysregulated transcriptional programs in hematopoietic stem/progenitor cells (HSPCs)** → **enhanced HSC self-renewal and myeloid-lineage skewing** (branch toward pre-malignancy) [PMID: 26256876, 26256875, 32518946]. 4a. **Branch — malignant predisposition:** aberrant self-renewal creates a **pre-malignant HSPC state**; upon acquisition of inferred somatic "second hits," this **progresses to B-cell or T-cell lymphoma / myeloid neoplasia** [PMID: 26256875, 32518946]. 4b. **Branch — autoimmunity/lymphoproliferation:** loss of TET function in **FOXP3⁺ Treg cells** → increased methylation of intronic *FOXP3* enhancers (CNS2/TSDR) and **Treg instability** → **"ex-Treg" cells biased toward T follicular helper (Tfh) cells** → **expansion of Tfh and plasma cells with autoantibody production** [PMID: 41972131]. In parallel, **impaired Fas-dependent apoptosis** and **expanded double-negative T cells** produce an ALPS-like lymphoproliferative state [PMID: 32518946]. 4c. **Branch — humoral defect:** TET2 loss in B

cells → **defective class-switch recombination** → impaired antibody responses contributing to infection susceptibility [PMID: 32518946]. 4d. **Branch — inflammation (methylation-independent):** IκBζ targets TET2 to the *Il6* promoter, where TET2 recruits **HDAC2** to actively repress *IL6* during inflammation resolution; loss of TET2 → **failure to resolve inflammation, elevated IL-6**, endotoxin/colitis susceptibility [PMID: 26287468].

4. Convergence of these branches → the clinical syndrome: **infection susceptibility + chronic lymphadenopathy/hepatosplenomegaly + ALPS-like autoimmunity + chronic inflammation + lymphoma**, with **developmental delay** [PMID: 32518946].

Mechanistic diagram



Category detail

- **Molecular pathways:** DNA demethylation / 5mC→5hmC oxidation cycle; *FOXP3*/TSDR demethylation program in Tregs; IκBζ–TET2–HDAC2 axis at the *IL6* promoter; TOPD (TET-OGT-PROSER1-DBHS) regulatory complex [PMID: 42423046]. Suggested GO: GO:0080111

(DNA demethylation), GO:0006338 (chromatin remodeling), GO:0045589 (regulation of regulatory T cell differentiation), GO:0032088 (negative regulation of NF- κ B transcription factor activity), GO:0032715 (negative regulation of interleukin-6 production).

- **Cellular processes:** HSC self-renewal, myeloid differentiation, lymphocyte apoptosis (GO:0070227), class-switch recombination (GO:0045190), inflammation and its resolution.
- **Protein dysfunction:** Loss of function / absent protein; abolished catalytic 5-hydroxymethylation activity [PMID: 32518946].
- **Metabolic changes:** TET2 is Fe(II)/ α -ketoglutarate-dependent; its activity is linked to the TCA-cycle metabolite α -KG and to ascorbate as cofactor — a metabolic-epigenetic interface [PMID: 28823558, 42609618].
- **Immune system involvement:** Combined immunodeficiency + autoimmunity + lymphoproliferation + chronic inflammation — a "bridge between cancer and immunity" [PMID: 36066697].
- **Tissue damage / inflammatory mechanisms:** IL-6-driven inflammation, ALPS-like lymphoproliferation, tissue infiltration (lymph nodes, liver, spleen, colon in models).
- **Epigenetic changes:** Genome-wide hypermethylation; reduced 5hmC; locus-specific *FOXP3* enhancer hypermethylation.
- **Molecular profiling:** Whole-blood DNA hypermethylation (methylation profiling); iPSC-derived hematopoietic assays showing myeloid skew [PMID: 32518946]; mouse 5hmC/transcriptomic changes [PMID: 26256876, 26256875].
- **Advanced technologies:** Patient-derived iPSC hematopoietic differentiation demonstrated cell-intrinsic myeloid skewing; reversible RNAi mouse models demonstrated reversibility of the self-renewal phenotype on TET2 restoration [PMID: 28823558].

Cell types (CL suggestions): hematopoietic stem cell (CL:0000037), regulatory T cell (CL:0000815), T follicular helper cell (CL:0002038), B cell (CL:0000236), macrophage (CL:0000235), dendritic cell (CL:0000451), common myeloid progenitor (CL:0000049).

7. Anatomical Structures Affected

Organ level. - Primary: bone marrow / hematopoietic system (UBERON:0002371 bone marrow; UBERON:0002390 hematopoietic system); immune system (UBERON:0002405). -

Lymphoid organs: lymph nodes (UBERON:0000029) — lymphadenopathy; spleen (UBERON:0002106) — splenomegaly; thymus (UBERON:0002370). - **Secondary organ**

involvement: liver (UBERON:0002107) — hepatomegaly; gut/colon in inflammatory models (UBERON:0001155); central nervous system inferred via developmental delay (UBERON:0001017). - **Body systems:** hematopoietic/immune (primary); hepatobiliary and lymphatic (secondary); nervous system (developmental delay).

Tissue and cell level. - Lymphoid and myeloid tissues; bone-marrow stroma also implicated — TET2 loss dysregulates bone-marrow mesenchymal stromal cells (BMSCs), altering their support of HSPCs and accelerating myeloid malignancy [PMID: 29290626]. - Cell populations: HSCs/HSPCs, Tregs, Tfh cells, double-negative T cells, B cells, macrophages, dendritic cells (see CL terms above).

Subcellular level (GO cellular component). Nucleus (GO:0005634) and chromatin (GO:0000785) — site of TET2 catalysis and the HDAC2 co-repressor complex.

Localization / lateralization. Systemic and bilateral (generalized lymphadenopathy, bilateral organ involvement); no lateralization.

8. Temporal Development

Onset. Congenital genetic lesion with childhood clinical onset; pattern is chronic/insidious, with progressive immune dysregulation and lymphoproliferation. Developmental delay indicates early-childhood impact.

Progression. Chronic and progressive, with episodic lymphoproliferative flares and autoimmune complications, evolving toward lymphoma. The malignancy step is a discrete, later event requiring inferred somatic second hits — consistent with a pre-malignant "gatekeeper" state that "creates a pre-malignant HSPC state requiring additional mutations for overt malignancy" [PMID: 26256875].

Disease course pattern. Progressive/lifelong without curative intervention; punctuated (episodic) autoimmune/lymphoproliferative events.

Patterns / remission. No spontaneous remission of the underlying defect. **Treatment-induced remission** is achieved by allogeneic HSCT, which corrects the hematopoietic-immune compartment. **Critical window:** early diagnosis and transplantation before malignant transformation represents the key opportunity for intervention.

9. Inheritance and Population

Epidemiology. Ultra-rare; only a small number of families reported worldwide. No reliable prevalence or incidence estimate exists; it is best described as <1 in 1,000,000 by extrapolation from the handful of cases.

Inheritance. Autosomal recessive — "the first reported cases of autosomal-recessive germline TET2 deficiency in humans" [PMID: 32518946]. Biallelic (homozygous or compound-heterozygous) LOF is required for the full syndrome; **heterozygous** carriers show a partial/attenuated phenotype with lymphoma and autoimmune features [PMID: 40031954].

Penetrance / expressivity. Biallelic LOF appears highly penetrant for immune dysregulation and lymphoma predisposition in reported cases (small n). Expressivity is variable (B-cell vs T-cell lymphoma; variable autoimmunity; impaired Fas apoptosis in 2 of 3). Heterozygous variants show **reduced penetrance / age-dependent expressivity**.

Genetic anticipation. Not applicable (not a repeat-expansion disorder).

Germline mosaicism / founder effects. Not reported. **Consanguinity** contributes to homozygosity in autosomal-recessive inheritance and is a relevant counseling consideration.

Carrier frequency. Rare individual LOF alleles; no established population carrier frequency for disease-causing biallelic states. (Somatic TET2 mutations in clonal hematopoiesis are common but etiologically distinct.)

Population demographics. No specific ethnic predilection established given the small case count. Both sexes affected (no established sex bias). Age distribution: childhood-onset.

10. Diagnostics

Clinical / laboratory tests. - CBC with cytopenias; immunoglobulin levels (hypergammaglobulinemia reported); lymphocyte subset immunophenotyping showing **expanded double-negative TCR $\alpha\beta$ +CD4–CD8– T cells**, depleted circulating Tfh, altered DC ratios [PMID: 32518946, 36066697]. - **ALPS-like biomarkers:** elevated **vitamin B12**, elevated **IL-10**, elevated double-negative T cells [PMID: 36066697]. - **Functional assays:** Fas-mediated apoptosis assay (impaired in a subset); B-cell class-switch recombination assay (defective) [PMID: 32518946]. - **Epigenetic assay:** global DNA methylation / 5hmC quantification showing whole-blood hypermethylation and reduced 5hmC [PMID: 32518946, 36066697].

Genetic testing. - **Definitive test: whole-exome or whole-genome sequencing** identifying biallelic germline *TET2* LOF variants (the discovery method) [PMID: 32518946]. Targeted *TET2* single-gene testing or inclusion in **primary immunodeficiency / ALPS / bone-marrow-failure gene panels** is appropriate once suspected. - Confirm **germline** origin (versus somatic) using non-hematopoietic tissue (e.g., fibroblasts) and parental segregation — clinically important because *TET2* is a common somatic driver. "Assessment for *TET2* mutations and germline origin should be considered in the appropriate context" [PMID: 40031954]. - Chromosomal microarray/karyotype not diagnostic for the germline lesion; useful in workup of associated malignancy.

Omics-based diagnostics. Genome-wide methylation profiling provides a functional biomarker of *TET2* loss; functional 5hmC quantification supports variant interpretation.

Clinical criteria / differential diagnosis. No formal consensus criteria yet. Differential diagnosis includes **ALPS (FAS/FASLG/CASP10)**, other combined immunodeficiencies with autoimmunity (e.g., CTLA4, LRBA, STAT3 GOF), and inherited bone-marrow-failure/lymphoma-predisposition syndromes. Distinguishing features of IMD79: biallelic germline *TET2* LOF, global DNA hypermethylation, myeloid-skewed hematopoiesis, and B-/T-cell lymphoma predisposition.

Screening. Cascade genetic testing of relatives once a familial variant is identified; carrier testing in consanguineous families.

11. Outcome / Prognosis

Survival / mortality. Formal survival statistics are unavailable (tiny cohorts). Prognosis is driven by **lymphoma** and infection/autoimmune complications. Untreated, the natural history is progressive with high malignancy risk. **All three index patients achieved early autologous T-cell reconstitution after allogeneic HSCT** [PMID: 32518946], indicating that HSCT can correct the hematopoietic-immune disease and is potentially curative.

Morbidity / function. High morbidity from recurrent infection, chronic lymphoproliferation, autoimmune complications, developmental delay, and cancer therapy/transplant. Formal QoL instruments have not been applied.

Complications. B-cell and T-cell lymphoma (major); autoimmune cytopenias and multi-system autoimmunity; chronic inflammation (IL-6-driven); transplant-related morbidity.

Prognostic factors. Development of lymphoma is the principal adverse prognostic event; earlier transplantation before malignant transformation is favorable. Zygosity (biallelic vs heterozygous) and residual TET2 activity likely modulate severity.

12. Treatment

Definitive / curative therapy. - Allogeneic hematopoietic stem cell transplantation (HSCT) — corrects the cell-intrinsic hematopoietic-immune defect; all three index patients transplanted with early autologous T-cell reconstitution [PMID: 32518946]. NCIT suggestion: NCIT:C15431 (Hematopoietic Stem Cell Transplantation) / NCIT:C15409 (Allogeneic Bone Marrow Transplantation).

Pharmacotherapy / supportive. - Immunosuppression / immunomodulation for autoimmune manifestations (as in ALPS management: corticosteroids, sirolimus, mycophenolate) — extrapolated, disease-specific data limited. - Anti-infective prophylaxis and immunoglobulin replacement for the immunodeficiency component (supportive standard of care). - Lymphoma-directed chemotherapy/immunotherapy for malignant transformation, per histology.

Mechanism-based investigational adjunct — Vitamin C (ascorbate). - Rationale: TET2 is an Fe(II)/ α -KG dioxygenase for which ascorbate is a cofactor. In a reversible RNAi model, "**Tet2 restoration reverses aberrant hematopoietic stem and progenitor cell (HSPC) self-renewal in vitro and in vivo,**" and vitamin C mimics restoration by acting as a TET cofactor [PMID: 28823558]. Ascorbate "inhibits proliferation and promotes myeloid differentiation" in TET2 loss-of-function contexts [PMID: 34497762]. - **Caveat:** This strategy requires **residual or paralog (TET1/TET3) activity** to act upon; in complete biallelic null IMD79 with absent protein, ascorbate cannot restore activity. It is therefore a mechanistically rational but **unproven** adjunct, most plausibly relevant to hypomorphic alleles. CHEBI: CHEBI:38290 (L-ascorbate) / NCIT:C285 (Ascorbic Acid).

Advanced / future therapeutics. Gene therapy / gene correction of *TET2* is conceptually attractive but not clinically available. Targeting the downstream inflammatory arm (e.g., **IL-6 pathway blockade**, tocilizumab; NCIT:C64485) is mechanistically supported by the TET2–HDAC2–*IL6* axis [PMID: 26287468], though unproven in IMD79.

Treatment strategy. Personalized: confirm germline zygosity and residual function; manage autoimmunity/infection supportively; monitor for malignancy; proceed to allogeneic HSCT as definitive therapy, ideally before malignant transformation.

13. Prevention

- **Primary prevention:** Not possible for a monogenic germline disease. **Genetic counseling** for consanguineous or carrier families, with options for **preimplantation genetic diagnosis / prenatal testing** where a familial variant is known.
- **Secondary prevention:** **Cascade genetic testing** and **surveillance** of biallelic (and heterozygous) carriers for autoimmune manifestations and lymphoma, enabling early intervention.
- **Tertiary prevention:** Aggressive management of autoimmunity, infection prophylaxis, and cancer surveillance to prevent complications; timely HSCT to pre-empt malignant transformation.
- **Immunization / public health / environmental:** Standard immunization and infection-avoidance measures for immunodeficient patients (with attention to live-vaccine risks); no vector/sanitation interventions relevant.
- **Counseling:** Genetic counseling emphasizing autosomal-recessive inheritance, consanguinity risk, distinction from somatic *TET2* mutations, and reduced-penetrance risk in heterozygotes.

14. Other Species / Natural Disease

- **Taxonomy / orthologs:** *TET2* is conserved in mammals. Mouse *Tet2* (NCBI Gene 214133) is the principal experimental ortholog. NCBI Taxon: *Mus musculus* (10090).
- **Natural disease in other species:** No well-characterized spontaneous germline TET2-deficiency disease entity is established in companion animals or wildlife (no OMIA entry identified). Somatic TET2-related myeloid pathology is chiefly a human/rodent-model concern.
- **Comparative biology / conservation:** Disease mechanisms are evolutionarily conserved — mouse *Tet2* loss recapitulates enhanced HSC self-renewal, myeloid skewing, MDS/MPN-like disease, and T-cell lymphoma [PMID: 26256876, 26256875], and the IL-6/inflammation-resolution role is conserved in mouse innate immune cells [PMID: 26287468]. TET1 is non-redundant (dispensable for MLL-ENL leukemogenesis) [PMID: 33705482].
- **Transmission / zoonosis:** Not applicable (non-infectious genetic disease).

15. Model Organisms

Mouse models (principal system).

Model	Key phenotype relevant to IMD79	PMID
Tet2 knockdown (RNAi)	↑ serial replating; ↑ HSC self-renewal in competitive repopulation & serial transplantation	26256876
Tet2 knockout / mutant	Enhanced self-renewal; MPN/MDS-like disease and T-cell lymphoma	26256875
Tet2(trap/trap)	High early-postnatal lethality — TET2 essential for survival/HSC homeostasis	26256875
Reversible RNAi (Tet2 restoration)	Restoration reverses aberrant HSPC self-renewal in vitro and in vivo ; vitamin C mimics restoration	28823558
Tet2-deficient (inflammation)	↑ susceptibility to endotoxin shock and DSS colitis; ↑ IL-6; failure to resolve inflammation	26287468
Tet2 loss in BMSCs	Altered stroma promotes Tet2-deficiency-mediated myeloid malignancy progression	29290626
Treg-specific TET loss	Ex-Treg → Tfh skewing; Tfh/plasma-cell expansion; autoantibody-mediated autoimmunity	41972131
Tet2-mutated myeloid progenitors (in vitro)	Aberrant in vitro self-renewal capacity	24786459

Human cellular models. Patient-derived iPSCs: "The hematopoietic potential of patient-derived induced pluripotent stem cells was skewed toward the myeloid lineage" [PMID: 32518946] — a humanized in vitro model capturing the cell-intrinsic hematopoietic defect.

Model characteristics. Mouse models faithfully recapitulate the hematopoietic self-renewal, myeloid skewing, malignancy predisposition, Treg/autoimmunity, and inflammation-resolution phenotypes — i.e., most mechanistic arms of IMD79. **Limitations:** the full combined-immunodeficiency + developmental-delay clinical picture of the human germline biallelic syndrome is not comprehensively modeled; second-hit requirements and lymphoma latency complicate direct correspondence; conditional/tissue-specific models capture individual arms rather than the integrated disease.

Resources. MGI (mouse *Tet2*), IMPC/KOMP knockout lines, and patient iPSC lines.

Key Findings (with evidence)

Finding 1 — IMD79 is caused by biallelic germline loss-of-function *TET2* mutations

Whole-exome sequencing of three unrelated children with immune dysregulation identified rare homozygous germline missense or nonsense *TET2* variants at 4q24; the mutant protein was absent or enzymatically dead for 5-hydroxymethylation, causing whole-blood DNA hypermethylation. This defined the first cases of **autosomal-recessive germline *TET2* deficiency** (OMIM #619223). The entity was expanded by a compound-heterozygous ALPS-like patient [PMID: 36066697] and four heterozygous LOF carriers with B-cell lymphoma [PMID: 40031954].

"we identified rare homozygous germline missense or nonsense variants in a known epigenetic regulator of gene expression: ten-eleven translocation methylcytosine dioxygenase 2 (*TET2*). Mutated *TET2* protein was absent or enzymatically defective for 5-hydroxymethylating activity, resulting in whole-blood DNA hypermethylation" — PMID: [32518946](#)

Finding 2 — Clinical phenotype: infection susceptibility, lymphoproliferation, autoimmunity, lymphoma

Three index children presented with infection susceptibility, lymphadenopathy, hepatosplenomegaly, developmental delay, autoimmunity, and B-cell (n=2) or T-cell (n=1) lymphoma, with expanded double-negative T cells, depleted Tfh, impaired Fas apoptosis (2/3), and defective B-cell class-switch recombination. All three achieved early autologous T-cell reconstitution after allogeneic HSCT.

"an immune dysregulation syndrome of susceptibility to infection, lymphadenopathy, hepatosplenomegaly, developmental delay, autoimmunity, and lymphoma of B-cell (n = 2) or T-cell (n = 1) origin" — PMID: [32518946](#)

Finding 3 — TET2 loss skews hematopoiesis and drives HSC self-renewal

Patient iPSCs skew toward myeloid lineage; mouse Tet2 loss reduces marrow 5hmC, increases serial replating and HSC self-renewal/competitive repopulation, and produces MDS/MPN-like disease and T-cell lymphoma; Tet2(trap/trap) mice show high early lethality. TET2 acts as a "gatekeeper" whose loss creates a pre-malignant state requiring additional mutations.

"TET2 knockdown led to an increased serial replating capacity of BM cells in vitro and increased hematopoietic stem cell (HSC) self-renewal in vivo in competitive repopulation and serial transplantation assays" — [PMID: 26256876](#) "Tet2 mutations induce enhanced self-renewal ability and competitive repopulation capacity in hematopoietic stem cells, and various MPN/MDS-like diseases and T-cell lymphoma consequently develop in model mice" — [PMID: 26256875](#)

Finding 4 — TET2 loss in Tregs promotes ex-Treg/Tfh conversion and autoimmunity

Loss of TET function in FOXP3⁺ Tregs yields "ex-Treg" cells biased toward Tfh, with expansion of Tfh and plasma cells and autoantibody-driven autoimmunity — a direct mechanism for the ALPS-like autoimmune arm.

"Loss of TET function in T regulatory cells yields ex-Treg cells biased toward T follicular helper cells, causing autoimmune diseases through autoantibody production" — [PMID: 41972131](#)

Finding 5 — Vitamin C is a mechanistically rational TET-restoration adjunct

TET2 restoration reverses aberrant HSPC self-renewal, and vitamin C mimics restoration as a cofactor of Fe(II)/ α -KG TET enzymes; ascorbate inhibits proliferation and promotes myeloid differentiation in TET2-LOF contexts. Rational but unproven for complete germline null IMD79.

"Tet2 restoration reverses aberrant hematopoietic stem and progenitor cell (HSPC) self-renewal in vitro and in vivo" — [PMID: 28823558](#)

Finding 6 — TET2 restrains IL-6 via HDAC2 recruitment (methylation-independent)

TET2 actively represses *IL6* during inflammation resolution by recruiting HDAC2 (independently of DNA methylation); Tet2-deficient mice are more susceptible to endotoxin shock and DSS colitis with increased IL-6 — the inflammatory arm of IMD79.

"Tet2 selectively mediates active repression of interleukin-6 (IL-6) transcription during inflammation resolution in innate myeloid cells, including dendritic cells and macrophages" — [PMID: 26287468](#) "Tet2-deficient mice were more susceptible to endotoxin shock and dextran-sulfate-sodium-induced colitis, displaying a more severe inflammatory phenotype and increased IL-6 production compared to wild-type mice" — [PMID: 26287468](#)

Mechanistic Model / Interpretation

IMD79 is unified by a single molecular lesion — **loss of TET2's ability to convert 5mC to 5hmC** — acting through **two mechanistic arms** that jointly explain an unusually broad clinical picture:

1. **A catalytic/epigenetic arm** driven by genome-wide DNA hypermethylation. This deranges the transcriptional programs of HSCs (enhanced self-renewal, myeloid skew → pre-malignant "gatekeeper" state → lymphoma with second hits), Tregs (FOXP3 enhancer hypermethylation → ex-Treg→Tfh conversion → autoantibody autoimmunity), and B cells (defective class-switch recombination → antibody deficiency). This arm accounts for the immunodeficiency, ALPS-like autoimmunity, and cancer predisposition.
2. **A methylation-independent arm** in which TET2 normally recruits HDAC2 to the *IL6* promoter to shut off IL-6 during inflammation resolution. Its loss produces exaggerated, poorly resolving IL-6-driven inflammation — explaining the chronic inflammatory tone and heightened response to infectious/endotoxic triggers.

The model is strongly supported by convergent human (index cases, iPSCs) and mouse evidence, and importantly is **partially reversible**: restoring TET2 (or supplying its cofactor vitamin C where residual activity exists) reverses the aberrant HSPC self-renewal, and IL-6 pathway blockade is a rational target for the inflammatory arm. Definitive therapy, however, remains **allogeneic HSCT**, which replaces the defective hematopoietic-immune compartment.

Evidence Base

PMID	Title (abbrev.)	Role	Evidence type
32518946	Germline TET2 LOF causes childhood immunodeficiency and lymphoma	Defining paper — gene, variants, phenotype, iPSC, HSCT	Human clinical + iPSC
36066697	Novel germline TET2 mutations, ALPS-like + malignancy	Expands entity; ALPS biomarkers; absent TET2/hypermethylation	Human clinical
40031954	Heterozygous germline TET2 LOF, ALPS-like	Extends risk to heterozygous state; lymphoma	Human clinical
41972131	TET loss in Tregs → ex-Treg/Tfh	Autoimmune mechanism	Mouse
26256876	Tet2 knockdown mouse	HSC self-renewal ↑	Mouse
26256875	TET2 as gatekeeper	Self-renewal, MPN/MDS, T-cell lymphoma	Mouse review
28823558	Restoration of TET2 blocks aberrant self-renewal	Reversibility; vitamin C rationale	Mouse
34497762	Ascorbate promotes myeloid differentiation in TET2-LOF	Vitamin C adjunct rationale	In vitro
26287468	Tet2 recruits Hdac2 to repress IL-6	Inflammatory arm	Mouse
29290626	TET2 loss dysregulates BMSCs	Stromal contribution	Mouse
24786459	Tet2-mutated myeloid progenitors self-renew	In vitro self-renewal	In vitro
33705482	Tet1 dispensable for MLL-ENL leukemogenesis	TET member non-redundancy	Mouse
42423046	TOPD complex regulation of TET	Non-catalytic regulation	Review
42609618	TET2 as therapeutic hub	Cofactor biology, breadth	Review

Limitations and Knowledge Gaps

- **Tiny cohort.** The disease is defined from very few patients; prevalence, penetrance, expressivity, survival, and QoL cannot be quantified reliably.
 - **Second-hit uncertainty.** The specific somatic events driving progression from pre-malignant HSPC state to overt lymphoma in IMD79 patients are inferred, not systematically characterized.
 - **Vitamin C evidence gap.** Ascorbate's benefit is demonstrated in TET2-hypomorphic/somatic models, not in germline null IMD79; efficacy in complete biallelic loss is mechanistically implausible and clinically untested.
 - **Treatment evidence.** Beyond HSCT (n=3), immunosuppressive/anti-IL-6 strategies are extrapolated from ALPS and mouse data, not validated in IMD79.
 - **Ontology mapping.** A dedicated Orphanet code and a fully mapped MONDO/ICD-11 entry are not yet established for this ultra-rare disease.
 - **Genotype–phenotype.** How specific missense vs null alleles, and residual TET1/TET3 activity, modulate severity is unresolved.
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Proposed Follow-up Experiments / Actions

1. **International patient registry & natural-history study** for biallelic and heterozygous germline *TET2* carriers to quantify penetrance, lymphoma risk, and HSCT outcomes.
2. **Systematic somatic genomics** of IMD79-associated lymphomas to define the required second hits and inform surveillance/therapy.
3. **Genotype–function correlation:** measure residual 5hmC/enzymatic activity for each variant and correlate with severity; test which alleles are ascorbate-responsive.
4. **Preclinical/early-phase test of the inflammatory arm:** evaluate IL-6 pathway blockade (e.g., tocilizumab) for the chronic inflammatory manifestations, guided by the TET2–HDAC2–*IL6* mechanism.
5. **Ascorbate pharmacodynamic study** in patients/cells with hypomorphic alleles, using 5hmC restoration and myeloid differentiation as biomarkers.

6. **Optimize HSCT timing** — define pre-transplant surveillance to intervene before malignant transformation.
 7. **Formal ontology curation:** establish MONDO/Orphanet/ICD-11 mappings and HPO annotation set with frequencies for the disease knowledge base.
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Report compiled from a 5-iteration autonomous literature investigation; 6 confirmed findings; 17 papers reviewed. Evidence types span human clinical case series, patient-derived iPSC models, mouse genetic models, and in vitro assays.

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